

```
!pip install transformers datasets torch scikit-learn pandas matplotlib
```

```

ing>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
zers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
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-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
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=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
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poolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
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t_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (
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3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2.5.0)
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<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Safe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
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=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer-

```

```
import torch
import pandas as pd
import numpy as np

from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from transformers import Trainer, TrainingArguments

from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

```
dataset = load_dataset("imdb")
print(dataset)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets
  warnings.warn(
README.md:      7.81k/? [00:00<00:00, 520kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable faster authentication.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub
plain_text/train-00000-of-100000.parquet: 21.0M/21.0M [00:00<00:00, 29.4MB/s]
00001.parquet: 100%
plain_text/test-00000-of-100000.parquet: 20.5M/20.5M [00:02<00:00, 33.5MB/s]
00001.parquet: 100%
plain_text/unsupervised-00000-of-100000.parquet: 42.0M/42.0M [00:01<00:00, 79.9MB/s]
00001.p(...): 100%
Generating train split: 100%                25000/25000 [00:00<00:00, 73120.95 examples/s]
Generating test split: 100%                 25000/25000 [00:00<00:00, 109939.41 examples/s]
Generating unsupervised split: 100%         50000/50000 [00:00<00:00, 130442.34 examples/s]
DatasetDict({
  train: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  test: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  unsupervised: Dataset({
    features: ['text', 'label'],
    num_rows: 50000
  })
})
```

```
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
config.json: 100%                483/483 [00:00<00:00, 11.9kB/s]
tokenizer_config.json: 100%       48.0/48.0 [00:00<00:00, 1.12kB/s]
vocab.txt:      232k/? [00:00<00:00, 6.78MB/s]
tokenizer.json:  466k/? [00:00<00:00, 10.5MB/s]
```

```
def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)
```

```
tokenized_datasets = dataset.map(tokenize_function, batched=True)
```

```

Map: 100%                               25000/25000 [00:31<00:00, 730.90 examples/s]
Map: 100%                               25000/25000 [00:30<00:00, 857.54 examples/s]
Map: 100%                               50000/50000 [01:03<00:00, 672.69 examples/s]

```

```

tokens = tokenizer("This movie is amazing!")
print(tokens)

```

```

{'input_ids': [101, 2023, 3185, 2003, 6429, 999, 102], 'token_type_ids': [0, 0, 0, 0, 0, 0, 0], 'a

```

```

train_dataset = tokenized_datasets["train"]
test_dataset = tokenized_datasets["test"]

```

```

train_dataset.set_format(type="torch", columns=["input_ids", "attention_mask", "label"])
test_dataset.set_format(type="torch", columns=["input_ids", "attention_mask", "label"])

```

```

model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)

```

```

model.safetensors: 96%                               256M/268M [00:06<00:00, 210MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 381.69it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.w

```

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased

Key	Status
-----+-----	-----+-----
vocab_layer_norm.weight	UNEXPECTED
vocab_transform.weight	UNEXPECTED
vocab_projector.bias	UNEXPECTED
vocab_layer_norm.bias	UNEXPECTED
vocab_transform.bias	UNEXPECTED
classifier.weight	MISSING
pre_classifier.weight	MISSING
classifier.bias	MISSING
pre_classifier.bias	MISSING

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect
- MISSING :those params were newly initialized because missing from the checkpoint. Consider tra

```

training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    per_device_eval_batch_size=16,
    num_train_epochs=2,
    weight_decay=0.01,
)

```

```

def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)

    precision, recall, f1, _ = precision_recall_fscore_support(labels, predictions, average='binary')
    acc = accuracy_score(labels, predictions)

    return {
        "accuracy": acc,
        "f1": f1,

```

```
"precision": precision,  
"recall": recall  
}
```

```
trainer = Trainer(  
    model=model,  
    args=training_args,  
    train_dataset=train_dataset,  
    eval_dataset=test_dataset,  
    compute_metrics=compute_metrics  
)
```

```
trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' is deprecated. Please use 'pin_memory_device' instead.  
  super().__init__(loader)
```

 [66/3126 55:49 < 44:29:13, 0.02 it/s, Epoch 0.04/2]

Step Training Loss

 [68/3126 57:33 < 44:26:47, 0.02 it/s, Epoch 0.04/2]

Step Training Loss